

FUKUYAMA, Japan

WMO: 477670

Lat: 34.4469N

Lon: 133.2475E

Elev: 3

StdP: 101.29

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.6	-1.5	-7.7	2.0	3.1	-6.5	2.2	2.9	6.6	6.8	6.0	6.6	1.3	30	0.359

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.1	34.1	25.5	33.1	25.4	32.1	25.1	26.6	31.9	26.0	31.3	25.6	30.6	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.3	29.1	24.6	19.6	28.8	24.1	19.0	28.5	83.0	31.9	80.7	31.3	78.7	30.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.2	4.4	-4.6	35.6	1.4	1.1	-5.7	36.4	-6.5	37.1	-7.2	37.7	-8.3	38.5		
			-5.5	27.3	1.4	0.6	-6.5	27.7	-7.3	28.1	-8.1	28.4	-9.1	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.9	4.7	5.3	8.4	13.6	18.7	22.6	26.9	28.3	24.3	18.4	12.2	7.0	(6)
(7)		DBStd	8.58	2.19	2.63	2.93	3.12	2.34	2.24	2.14	1.82	2.83	2.93	3.14	2.66	(7)
(8)		HDD10.0	487	166	134	67	7	0	0	0	0	0	13	99		(8)
(9)		HDD18.3	1834	424	366	307	144	23	1	0	0	34	183	352		(9)
(10)		CDD10.0	2656	1	2	18	114	270	379	525	569	430	261	81	6	(10)
(11)		CDD18.3	962	0	0	0	2	36	129	267	311	180	37	1	0	(11)
(12)		CDH23.3	8895	0	0	0	13	201	771	2708	3589	1450	163	0	0	(12)
(13)		CDH26.7	3290	0	0	0	1	16	153	1068	1594	442	17	0	0	(13)
(14)	Wind	WSAvg	1.7	1.7	1.7	1.8	1.8	1.7	1.6	1.7	1.8	1.6	1.5	1.5	1.6	(14)
(15)	Precipitation	PrecAvg	1211	42	54	91	99	139	175	199	98	131	87	63	40	(15)
(16)		PrecMax	1863	152	100	206	173	309	384	511	266	258	241	147	97	(16)
(17)		PrecMin	726	7	18	44	30	31	24	36	16	45	25	12	2	(17)
(18)		PrecStd	268	31	25	38	37	58	89	135	72	59	65	39	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	16.1	20.0	25.0	28.4	31.8	34.8	35.7	33.6	28.6	22.2	16.7	(19)
(20)		0.4%	MCWB	9.2	11.7	12.9	16.2	18.8	24.0	25.7	25.7	25.1	22.8	16.9	11.8	(20)
(21)		2%	DB	12.0	13.6	17.8	22.9	26.7	29.7	33.5	34.6	31.9	26.5	20.8	15.1	(21)
(22)		2%	MCWB	7.6	9.3	11.8	15.1	18.2	22.6	25.6	25.6	24.6	20.5	15.8	10.8	(22)
(23)		5%	DB	10.7	12.1	16.0	21.2	25.5	28.4	32.6	33.6	30.6	25.1	19.4	13.7	(23)
(24)		5%	MCWB	6.7	8.0	10.8	14.6	17.6	21.8	25.4	25.5	24.0	19.1	14.9	9.7	(24)
(25)		10%	DB	9.6	10.6	14.3	19.7	24.3	27.2	31.6	32.7	29.2	23.8	17.9	12.3	(25)
(26)		10%	MCWB	5.8	6.8	9.8	13.7	17.2	21.2	25.1	25.4	23.0	18.4	14.0	8.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	12.8	14.6	18.9	21.5	25.0	27.4	27.2	26.1	23.7	18.6	13.9	(27)
(28)		0.4%	MCDWB	12.3	15.3	17.2	21.3	24.8	29.8	32.2	33.1	31.1	26.9	20.3	15.5	(28)
(29)		2%	WB	8.6	10.2	13.1	17.2	20.2	24.0	26.6	26.7	25.4	22.1	17.2	11.7	(29)
(30)		2%	MCDWB	10.8	12.6	15.8	20.2	23.5	28.1	31.6	32.3	29.9	25.5	19.5	14.1	(30)
(31)		5%	WB	7.4	8.8	11.9	16.0	19.4	23.2	26.0	26.2	24.8	20.6	15.9	10.3	(31)
(32)		5%	MCDWB	10.0	11.3	15.1	19.5	23.0	26.5	31.0	31.7	29.0	23.6	18.4	12.8	(32)
(33)		10%	WB	6.2	7.3	10.5	14.9	18.5	22.5	25.5	25.8	24.2	19.3	14.6	9.1	(33)
(34)		10%	MCDWB	9.1	10.1	13.9	18.5	22.6	25.6	30.3	31.0	28.1	22.6	17.4	11.9	(34)
(35)	Mean Daily Temperature Range		MDBR	9.0	9.3	10.0	10.2	9.9	7.8	7.4	8.1	8.2	9.2	9.5	9.1	(35)
(36)		5% DB	MCDBR	10.5	10.8	12.0	12.4	12.0	9.5	9.0	9.3	8.9	9.7	10.0	10.4	(36)
(37)			MCWBR	6.9	7.3	7.6	6.5	5.7	4.1	3.2	3.1	3.7	4.8	5.9	6.8	(37)
(38)			MCDBR	9.6	9.2	10.1	9.0	8.5	6.9	8.3	8.4	7.6	7.7	8.3	9.0	(38)
(39)		5% WB	MCWBR	7.0	6.9	7.3	5.6	5.0	3.4	3.2	3.2	3.3	4.5	5.8	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.411	0.456	0.520	0.549	0.550	0.569	0.569	0.539	0.495	0.450	0.425	0.404	(40)	
(41)		taud	2.108	2.010	1.865	1.850	1.912	1.940	1.994	2.071	2.157	2.192	2.159	2.145	(41)	
(42)		Ebn at Noon	763	770	753	755	760	743	740	756	770	771	744	743	(42)	
(43)		Edh at Noon	125	153	192	204	194	188	178	162	142	126	118	114	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.33	2.95	3.95	4.84	5.34	4.74	5.04	5.17	4.08	3.40	2.52	2.04	(44)	
(45)		RadStd	0.16	0.24	0.30	0.43	0.50	0.45	0.66	0.51	0.42	0.33	0.21	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+0.32	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page